

NH₃ hyperfine intensity anomalies

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Summary. A search for intensity anomalies within the hyperfine structure of the NH₃ (J, K) = (1, 1) transition was performed. Out of 23 selected sources (associated with H₂O-maser and IR-emission) in the rotational temperature regime between T_R = 15 and 30 K, 13 showed non LTE behaviour. Thus, nonthermal population of the (1, 1) hyperfine states seems to be common in intermediate warm clouds, although no obvious correlation with other known properties of the clouds has emerged.

Key words: molecular clouds – molecules

I. Introduction

NH₃-emission has been used effectively to determine the gas kinetic temperature of dense cores in interstellar clouds. Knowledge of the optical depth is necessary for deriving the temperature from the intensity for the (J, K) = (1, 1) and (2, 2) lines. In deriving the optical depth from a comparison of the intensities of the nuclear quadrupole hyperfine structure (hfs) components with the main line, it is generally assumed that the hfs-levels have equal excitation temperature.

For an increasing number of clouds however, the hfs-levels exhibit a nonthermal population resulting in deviations of the hfs-line intensities from their equilibrium values. Such anomalous intensities were first discovered in absorption towards the continuum source DR 21 (Matsakis et al., 1977). Subsequently it was found for several other sources such as S106 (Stutzki et al., 1982), OMC2 (Bartla et al., 1983), and W48 (Morane, 1984).

Matsakis et al., 1977, proposed a mechanism to explain the observed anomalies, which relies on the NH₃-gas being clumped. The individual clumps are assumed to possess very small line widths (~ 0.3 km s⁻¹), whereas the observed larger line widths must result from the relative motion of the clumps. The anomaly is then produced by selective trapping in the hyperfine structure of the (2, 1) $\rightarrow$ (1, 1) rotational transition.

Radiative transfer calculations based on this mechanism and taking into account the proper hyperfine selective collisional excitation, reproduce anomalies of the same kind as the observed ones; then it is possible to determine not only kinetic temperature and column density, but, from the strength of the anomalies, also the H₂-densities, solely from the relative intensities of the (1, 1) hyperfine transitions and the (2, 2) main line. The details of this

calculations and the interpretation of the anomalies will be given in a separate paper (Stutzki and Winnewisser, 1984; see also Stutzki, 1984).

We undertook a systematic high signal to noise ratio search to obtain precise data on the anomalous hyperfine intensities, its distribution, and possible correlation on other cloud parameters for a selected number of sources.

II. Experimental details and source selection

From all known NH₃ sources we selected for our search those which qualified for the following two criteria:

1. the estimated or measured rotational temperature lies in the range of 15–30 K, and
2. the line width (< 3.0 km s⁻¹) and velocity structure allow a clear separation of the hyperfine components.

Whereas the first criterium excludes all cold dark clouds (in which in spite of detailed observations no anomalies have been found), the second one removes many sources with hot cores in the immediate neighbourhood of the exciting object, e.g. in OMC1 the KL-nebula or the BN-object, and sources such as the galactic center where the hfs is blended by the velocity structure. From about 40 sources fulfilling both criteria we observed the 23 sources listed in Table 1 with a determination of the satellite/main line intensity ratio with an error of less than 3%.

The observations were carried out with the Effelsberg 100 m radiotelescope, (beamwidth 40" at the NH₃ inversion frequencies, maser receiver with $T_{\text{sys}} \sim 80$ K) and the Haystack Observatory¹ 37 m telescope (beam width 1.4', maser receiver with $T_{\text{sys}} \sim 90$ K). At Effelsberg the 384 channel autocorrelator was run in split mode (2×5 MHz, 192 channels each) for the simultaneous observation of the (1, 1) and (2, 2) line which results in a relative calibration more accurate than for the Haystack data, where the lines had to be observed separately (1024 channel autocorrelator, 6.67 MHz bandwidth).

To the calibrated spectra we fitted Gaussian line profiles for each quadrupole hfs component with a fixed separation of the hyperfine satellites [except for the spectra with two (or more) velocity components, where the fitting was done without fixed separations].

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III. Results

In a plot of the intensity ratio of the outer satellites of the (1, 1) line ($T_B(F_1=1 \rightarrow 0)/T_B(F_1=0 \rightarrow 1)$) (Fig. 1) the anomaly of the hyperfine components reveals itself as a deviation from the thermal equilibrium value of 1. In Table 2 we list all the sources showing anomalies, i.e. for which $T_B(F_1=1 \rightarrow 0)/T_B(F_1=0 \rightarrow 1)$ does not agree with 1 within 3σ . For these sources we list all measured relative intensities of the (1, 1) satellites. Included is the ratio of the

Table 1. List of sources

Source	Pos.	δ_{50}	v_{LSR} (km/s)	Ref.	telesc. c)
	α_{50}				
OMC2 a)	5 32 58.6	-5 11 42	10.0	5,6	E
OMC S1 a)	5 32 54.7	-5 14 04	10.0	6	E
S2 a)	5 32 53.5	-5 16 25	10.0	6	E
S3 a)	5 32 49.3	-5 21 02	10.0	6	E
S4 a)	5 32 48.0	-5 22 02	10.0	6	E
GGD4	5 37 22	23 49 24	2.3	3	H
S235AB	5 37 31.5	35 40 00	-17.0	1	H
Orion B	5 39 13	-1 55 48	9.0	2	H
G 205.11-14.11	5 44 31	0 20 48	8.85	3	H
Mon R2	6 05 19	-6 22 40	10.3	3	H
g Ophiuchi	16 23 25	-24 15 59	3.6	4	H
NGC 6334	17 17 32.5	-35 41 60	-4.0		H
OH 353.4-0.4	17 27 06.5	-34 39 28	-20.0		H
G 11.94-0.62	18 11 04.7	-18 54 29	37.3		H
W33	18 11 18.7	-17 56 40	35.0	2	H
G 24.49-0.04	18 33 23	-7 33 54	109.6	3	H
G 28.29-0.36	18 41 36.5	-4 21 00	49.0		H
G 32.15+0.13	18 46 58	-0 41 30	94.7	3	H
G 33.9+0.1	18 50 16	0 51 47	107.0	3	H
G 34.3+0.1	18 50 46	1 11 00	58.1	3	H
S76 E	18 53 47	7 49 26	32.7	3	H
W48 b)	18 59 12.8	1 08 12	42.7	3	E
S87	19 44 14	24 27 58	22.8	3	E
ON1	20 08 10	31 22 41	11.0	3	H
S106	20 25 25	37 12 30	-1.5	3,5,7	E
S106 OFF	20 19 37	37 12 30	-1.5		H
W75N	20 36 51	42 27.4	9.0	2	H

- a) data for these sources made available by Dr.W.Batrla
b) data for this source made available by A.Morane
c) H(E) means that the observations were done at Haystack (Effelsberg)
Ref.: 1) Ho et al., 1981
2) Schwartz et al., 1977
3) Macdonald et al., 1981
4) Myers et al., 1978
5) Little et al., 1980
6) Batrla et al., 1983
7) Stutzki et al., 1982

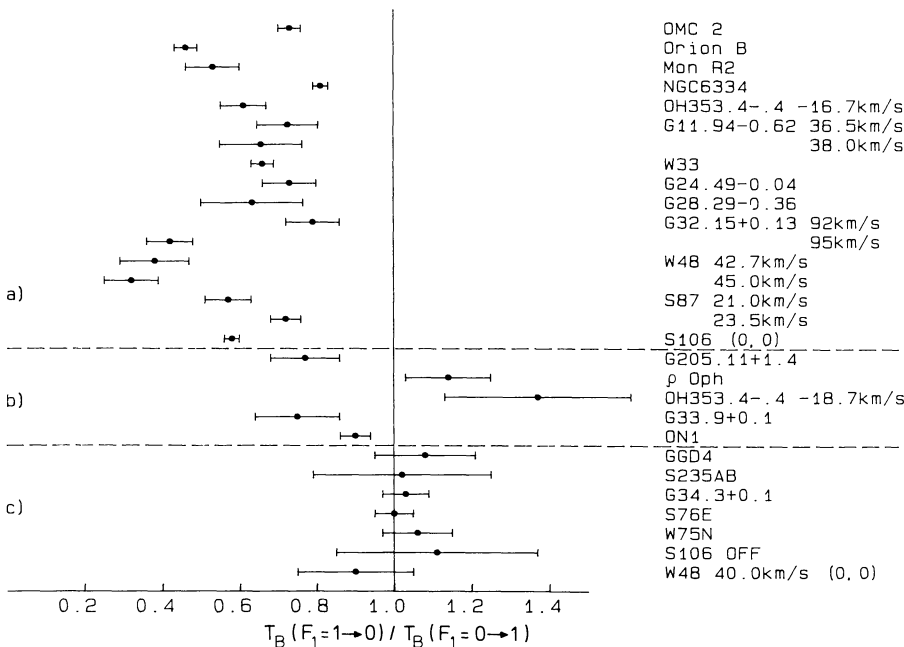


Fig. 1. For each source of Table 1 the intensity ratio of the outer satellites in the (1, 1) inversion transition, $T_B(F_1=1 \rightarrow 0)/T_B(F_1=0 \rightarrow 1)$ is plotted with 1σ error bars. The sources in a) deviate from the thermal equilibrium value of 1 by more than 3σ and therefore clearly show anomalous emission. The sources in b) only deviate from 1 less than 3σ and thus further observations are needed to confirm whether these sources are normal or not. The sources in c) agree with the thermal equilibrium value within 1σ . (For all spectra the signal to noise ratio is high enough to determine the satellite/main line intensity ratio with an error of 3%)

(2, 2) to the (1, 1) main line, the associated line width and LSR-velocities. For two of the sources (W48 and S106) mapping data are available. We include the results for these spectra in Table 2 [the data on S106 are somewhat improved compared to the ones published by Stutzki et al. (1982)]. Figure 2 shows the spectra of the anomalous sources.

Of the 13 sources listed in Table 2 (counting OMC2, OMCS1-S4 as one source), five show more than one velocity component: OH 353.4-0.4, G 11.94-0.62, G 32.15+0.13, W48, S87). In G 11.94-0.62, G 32.15+0.13 and S87 both components show anomalies, in W48 and OH 353.4-0.4 one of the components (also included in Table 1) shows normal emission. As the velocity separation of the two components is at least slightly larger than the linewidth, radiative interaction between the components seems not to be important.

The results for the remaining 10 sources and the components of W48 and OH 353.4-0.4 without anomalies are listed in Table 3. For these sources the outer satellite intensity ratio agrees with its thermal equilibrium value of 1 within 3σ . This does not, however, exclude that in some of these sources weak anomalies might be present; indeed five of the sources might be classified as showing anomalies, when one puts a 1σ instead of a 3σ limit (see Fig. 1) (two of them, however, have $T_B(1 \rightarrow 0)/T_B(0 \rightarrow 1) > 1$ in contrast to all the sources with definite anomalies, so that this probably is only an effect of poor signal to noise ratio). Table 3 also gives the value of τ_T derived from the fit of LTE-profiles to the hyperfine structure transitions (see Stutzki et al., 1982).

IV. Discussion

With 13 sources out of 23 showing anomalies it is clear that nonthermal population of the NH₃(1, 1) hyperfine structure levels is a fairly common phenomenon in the medium warm NH₃ clouds. The reason for their prior non-detection is that the high signal to noise ratio necessary for its identification was rarely achieved in earlier observations.

Table 2. Sources with hfs-anomalies

Source ($\Delta\alpha, \Delta\delta$) (")	$T_B(1,1)$ (K)	v_{LSR} (km/s)	Δv (km/s)	$R(F' \rightarrow F'') = T_B(F_1 = F' \rightarrow F'') / T_B(\Delta F_1 = 0)$				$T_B(2,2) / T_B(1,1)$
				$R(1 \rightarrow 0)$	$R(1 \rightarrow 2)$	$R(2 \rightarrow 1)$	$R(0 \rightarrow 1)$	
OMC 2 (d)	4.752(0.039)	11.2 (c)	1.259(0.012)	0.240(0.009)	0.325(0.009)	0.313(0.009)	0.329(0.009)	0.572(0.009)
OMC S1 (d)	9.413(0.042)	11.084(0.002)	0.944(0.016)	0.329(0.005)	0.409(0.005)	0.362(0.005)	0.365(0.005)	0.484(0.005)
S2 (d)	5.132(0.068)	10.797(0.006)	0.637(0.041)	0.281(0.016)	0.325(0.013)	0.367(0.014)	0.412(0.014)	0.518(0.013)
S3 (d)	7.115(0.049)	9.875(0.006)	1.506(0.053)	0.245(0.008)	0.372(0.007)	0.336(0.007)	0.377(0.008)	0.632(0.007)
S4 (d)	6.558(0.080)	9.698(0.008)	1.106(0.071)	0.248(0.014)	0.353(0.013)	0.325(0.013)	0.370(0.013)	0.693(0.012)
Orion B	1.020(0.024)	10.39(0.02)	2.00(0.05)	0.16(0.03)	0.29(0.02)	0.32(0.03)	0.35(0.03)	0.6 (a)
Mon R2	1.160(0.024)	10.73(0.02)	2.03(0.05)	0.19(0.02)	0.31(0.02)	0.30(0.03)	0.37(0.03)	0.3 (a)
NGC 6334	10.180(0.068)	-4.35(0.01)	3.55(0.03)	0.343(0.007)	0.469(0.008)	0.428(0.007)	0.424(0.007)	0.3 (b)
OH 353.4-0.4								
-18.7 km/s	2.20(0.28)	-18.70(0.06)	1.41(0.14)	0.70(0.10)	0.75(0.10)	0.50(0.09)	0.46(0.06)	(b)
-16.7 km/s	2.05(0.08)	-16.70(0.12)	2.78(0.26)	0.46(0.04)	0.66(0.07)	0.60(0.05)	0.74(0.04)	(b)
G11.94-0.62								
36.5 km/s	2.05(0.08)	36.54(0.02)	2.83(0.05)	0.39(0.04)	0.50(0.04)	0.50(0.04)	0.53(0.04)	(b)
38.3 km/s	1.80(0.07)	38.25(0.04)	3.31(0.08)	0.28(0.04)	0.45(0.04)	0.38(0.04)	0.43(0.04)	(b)
W33	1.967(0.016)	35.57(0.01)	3.45(0.03)	0.248(0.008)	0.361(0.008)	0.317(0.008)	0.378(0.009)	(b)
G24.49-0.04	1.06(0.02)	110.19(0.03)	3.67(0.09)	0.29(0.02)	0.46(0.02)	0.45(0.02)	0.42(0.02)	0.6 (a)
G28.29-0.36	1.32(0.02)	48.62(0.02)	2.78(0.05)	0.33(0.06)	0.58(0.07)	0.47(0.07)	0.53(0.07)	(b)
G32.15+0.13								
92 km/s	0.338(0.028)	91.83(0.07)	1.95(0.16)	0.48(0.10)	0.77(0.09)	0.72(0.09)	0.61(0.09)	0.4 (a)
95 km/s	0.996(0.024)	95.20(0.04)	3.30(0.11)	0.27(0.02)	0.51(0.03)	0.42(0.03)	0.51(0.03)	0.3 (a)
W48 (e)								
42.7 km/s -component								
-80, 40	1.56(0.06)	42.463(0.024)	1.59(0.07)	0.28(0.04)	0.35(0.04)	0.41(0.04)	0.42(0.04)	0.42(0.04)
-80, 80	2.60(0.05)	42.725(0.015)	1.72(0.04)	0.28(0.02)	0.40(0.02)	0.36(0.02)	0.34(0.02)	0.41(0.02)
-40, 40	1.40(0.04)	42.482(0.029)	2.66(0.09)	0.26(0.04)	0.48(0.03)	0.42(0.03)	0.44(0.03)	0.59(0.03)
-40, 80	4.12(0.05)	42.366(0.009)	2.01(0.03)	0.31(0.01)	0.48(0.01)	0.42(0.01)	0.48(0.01)	0.57(0.01)
0, 0	0.93(0.03)	42.749(0.036)	2.24(0.11)	0.24(0.04)	0.36(0.03)	0.32(0.04)	0.46(0.04)	0.43(0.03)
0, 40	0.98(0.05)	42.681(0.079)	1.86(0.16)	0.20(0.05)	0.36(0.06)	0.21(0.04)	0.55(0.06)	0.54(0.12)
0, 80	1.43(0.06)	42.394(0.032)	2.07(0.11)	0.27(0.05)	0.43(0.05)	0.41(0.05)	0.54(0.06)	0.63(0.04)
40, -40	1.33(0.07)	42.770(0.027)	1.28(0.08)	0.19(0.06)	0.34(0.06)	0.41(0.06)	0.41(0.06)	0.50(0.05)
80, -80	0.81(0.06)	42.713(0.061)	1.22(0.17)	0.34(0.08)	0.35(0.06)	0.38(0.10)	0.44(0.08)	0.40(0.06)
40.0 km/s -component								
-80, -40	1.58(0.08)	39.925(0.025)	1.24(0.07)	0.26(0.06)	0.36(0.05)	0.28(0.05)	0.30(0.05)	0.17(0.04)
-40, -40	1.86(0.08)	39.965(0.018)	1.50(0.06)	0.31(0.05)	0.46(0.05)	0.38(0.05)	0.33(0.05)	0.31(0.04)
0, 0	0.87(0.04)	39.798(0.027)	1.31(0.03)	0.34(0.05)	0.38(0.05)	0.35(0.06)	0.38(0.05)	0.29(0.06)
45.0 km/s -component								
0, 40	1.02(0.06)	44.725(0.062)	1.95(0.16)	0.24(0.05)	0.41(0.05)	0.32(0.06)	0.76(0.07)	0.67(0.07)
S87								
21.0 km/s	1.097(0.041)	20.88(0.03)	1.36(0.07)	0.32(0.05)	0.44(0.04)	0.44(0.05)	0.56(0.04)	0.63(0.04)
23.5 km/s	1.279(0.034)	23.49(0.03)	2.05(0.07)	0.28(0.03)	0.46(0.03)	0.31(0.03)	0.39(0.03)	0.69(0.03)
S106								
200, 80	1.32(0.09)	-1.95(0.04)	1.52(0.12)	0.39(0.08)	0.49(0.08)	0.42(0.07)	0.45(0.10)	0.48(0.10)
200, 40	3.04(0.06)	-2.23(0.01)	1.21(0.03)	0.32(0.02)	0.44(0.02)	0.34(0.02)	0.47(0.02)	0.51(0.05)
200, 0	1.08(0.13)	-1.61(0.06)	1.26(0.18)	0.24(0.11)	0.47(0.12)	0.40(0.10)	0.44(0.11)	0.36(0.10)
200, -40	1.53(0.12)	-1.96(0.05)	1.59(0.15)	0.35(0.08)	0.44(0.09)	0.40(0.08)	0.47(0.09)	0.33(0.07)
160, 80	1.50(0.12)	-1.66(0.07)	2.02(0.20)	0.28(0.07)	0.33(0.08)	0.24(0.08)	0.33(0.09)	0.48(0.08)
160, 40	2.49(0.08)	-2.12(0.02)	1.91(0.07)	0.27(0.03)	0.41(0.03)	0.34(0.03)	0.43(0.04)	0.55(0.04)
160, 0	1.96(0.04)	-1.49(0.01)	1.58(0.04)	0.23(0.02)	0.38(0.02)	0.37(0.02)	0.40(0.02)	0.46(0.02)
160, -40	1.94(0.12)	-2.13(0.03)	1.44(0.10)	0.38(0.06)	0.53(0.07)	0.42(0.06)	0.61(0.08)	0.43(0.06)
90, 30	0.55(0.03)	-1.42(0.05)	2.37(0.14)	0.22(0.06)	0.36(0.06)	0.27(0.04)	0.40(0.06)	0.60(0.06)
40, 40	0.94(0.10)	-0.97(0.07)	1.53(0.19)	0.20(0.14)	0.28(0.10)	0.37(0.11)	0.32(0.10)	0.56(0.11)
40, 0	0.84(0.03)	-1.29(0.02)	1.58(0.07)	0.20(0.05)	0.36(0.04)	0.40(0.05)	0.45(0.05)	0.60(0.05)
0, 40	1.38(0.03)	-1.17(0.02)	1.81(0.05)	0.26(0.02)	0.43(0.02)	0.40(0.02)	0.43(0.02)	0.50(0.02)
0, 0	4.713(0.060)	-1.336(0.007)	1.465(0.022)	0.294(0.014)	0.456(0.014)	0.428(0.014)	0.503(0.015)	0.52(0.02)
0, -40	1.42(0.12)	-0.75(0.06)	2.00(0.19)	0.20(0.09)	0.42(0.09)	0.42(0.10)	0.41(0.09)	0.42(0.07)
-40, 40	3.49(0.15)	-1.70(0.03)	1.72(0.09)	0.30(0.05)	0.58(0.05)	0.45(0.05)	0.57(0.05)	0.49(0.05)
-40, 0	3.57(0.04)	-1.32(0.01)	1.21(0.02)	0.30(0.01)	0.43(0.01)	0.45(0.01)	0.51(0.01)	0.45(0.01)
-40, -40	0.90(0.12)	-0.91(0.09)	1.79(0.27)	0.33(0.12)	0.49(0.13)	0.41(0.13)	0.32(0.11)	0.29(0.11)
-80, 40	1.90(0.09)	-1.83(0.03)	1.61(0.08)	0.29(0.05)	0.42(0.05)	0.42(0.04)	0.46(0.05)	0.39(0.05)
-80, 0	0.81(0.09)	-1.57(0.06)	1.28(0.17)	0.40(0.17)	0.25(0.10)	0.28(0.08)	0.38(0.12)	0.54(0.16)

- (a) relative calibration of (2,2)-line rather uncertain due to bad weather condition during observations of the (2,2)-line at Haystack
(b) (2,2) line not measured
(c) velocity scale uncertain due to problems with the equipment
(d) data on OMC2, OMC S1-S4 made available by Dr.W.Batrla
(e) data on W48 made available by A.Morane

Table 3. Sources without anomalies

Source	$T_B(1,1)$ (K)	v_{LSR} (km/s)	$\Delta v_{int}(a)$ (km/s)	τ_T
GGD4	0.96(0.03)	2.30(0.02)	1.22(0.06)	0.33(0.11)
S235AB	0.81(0.03)	-16.87(0.03)	1.94(0.08)	0.01(0.09)
G205.11+14	2.59(0.04)	9.31(0.01)	1.05(0.02)	0.31(0.04)
g Ophiuchi	3.00(0.06)	3.39(0.01)	0.89(0.02)	0.49(0.06)
OH353.4-0.4				
-16.7 km/s	2.68(0.08)	-16.83(0.07)	2.26(0.12)	1.73(0.15)
G33.9+0.1	1.13(0.03)	107.33(0.03)	2.54(0.06)	0.57(0.07)
G34.3+0.1	2.57(0.03)	58.20(0.02)	3.57(0.05)	0.39(0.04)
S76 E	1.34(0.02)	32.40(0.02)	3.07(0.05)	0.25(0.04)
W48 40.0 km/s	0.87(0.04)	39.82(0.03)	1.09(0.07)	0.55(0.16)
ON 1	1.87(0.02)	11.23(0.02)	2.84(0.04)	0.55(0.04)
S106 OFF	0.85(0.03)	-1.78(0.03)	1.83(0.08)	0.13(0.09)
W75 N	1.39(0.02)	9.36(0.02)	3.22(0.06)	0.25(0.04)

- (a) Δv_{int} is the intrinsic line width of an individual magnetic hfs-component (see Stutzki et al. 1982).

As shown in Fig. 1, most of the anomalous sources show an intensity ratio of the outer satellites $T_B(F_1 = 1 \rightarrow 0) / T_B(F_1 = 0 \rightarrow 1)$ in the interval 0.5–0.7. Some however have definitely lower values. The two classes of sources, i.e. with normal and anomalous hfs-behaviour are not well separated. This may not only be due to the observational errors, but may also be caused by a velocity blend of normal and anomalous spectra in some of the sources with only weak anomalies. Indeed, several of the sources investigated show two velocity components, one of them with anomalies, the other without. If two velocity components show only small separation, existing anomalies may be disguised.

We also note that in some sources, particularly NGC 6334, OMC, and W 48, the inner satellites appear to be non-thermal in the sense that $T_B(F_1 = 2 \rightarrow 1) / T_B(F_1 = 1 \rightarrow 2)$ is less than the theoretical thermal value of unity, as is also predicted by radiative transfer calculations (Stutzki, 1984; Stutzki and Winniewisser, 1984). Additional observations are required to confirm this result.

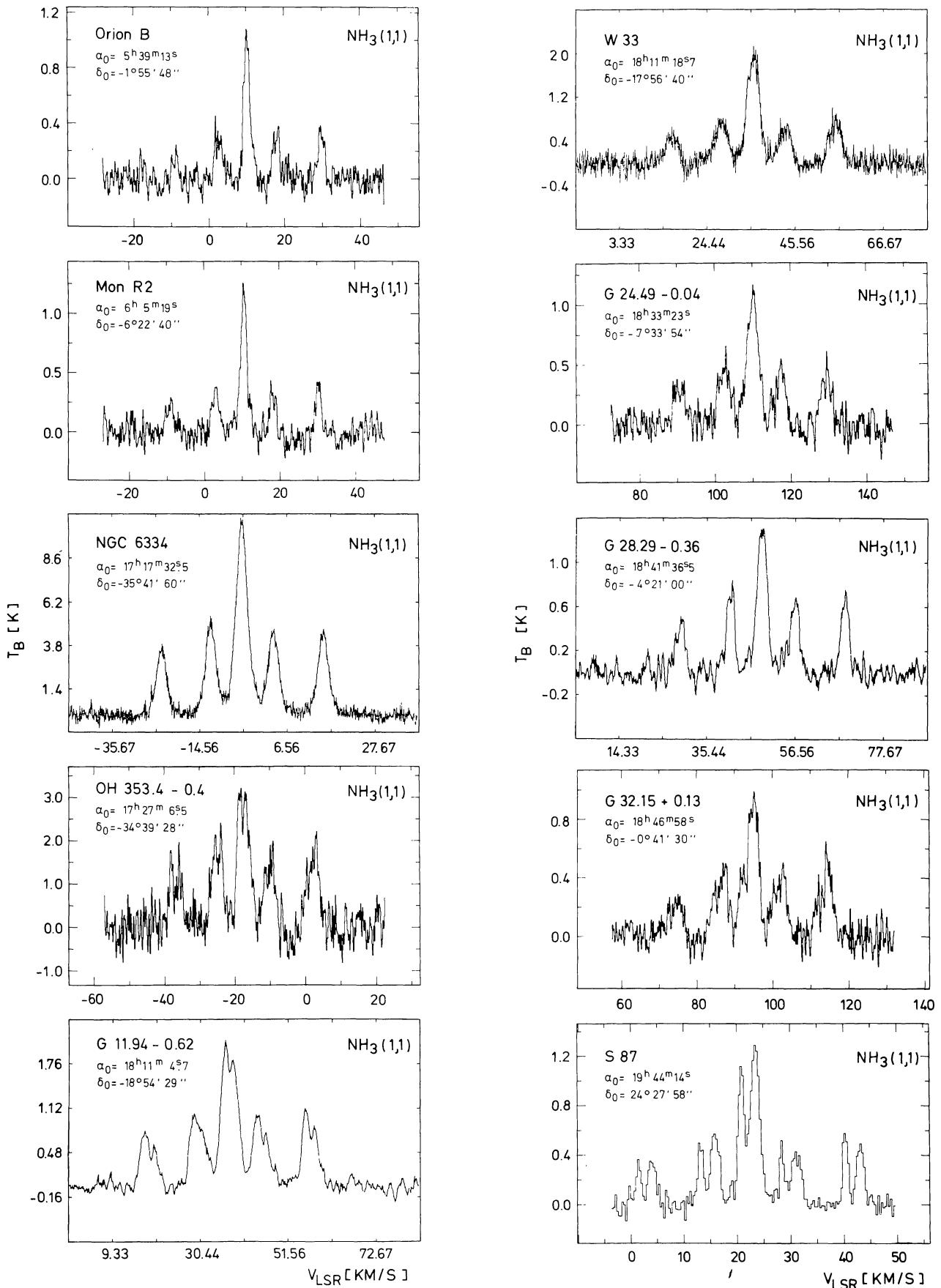


Fig. 2. The anomalous (1, 1) spectra of those sources, for which the spectra are not published elsewhere

In all sources with NH_3 hfs-anomalies, H_2O maser emission was found during our recent search between January 6 and 9, 1984, with the Haystack telescope. Since most warm clouds are associated with both H_2O maser and IR-emission sources, this coincidence between NH_3 anomalies and H_2O maser emission may just reflect the underlying selection criteria of our search. It is of interest to note, that the maser in S106, originally detected by Cesarsky et al. (1978), but not detected by Stutzki et al., 1982, was found to be active again with $T_B \sim 24$ K at -4.0 km s^{-1} on November 19 and $T_B \sim 8$ K on January 7. A second H_2O maser in S106, found by Stutzki et al., 1982, was also detected with a similar intensity.

Beside this coincidence with maser activity, there is no obvious property of the sources which seems to be correlated with the occurrence of hfs-anomalies. Our observations therefore do not indicate any special physical conditions which must be present for inducing nonthermal population in the $\text{NH}_3(1, 1)$ hyperfine states.

One of the basic assumptions of the usual analysis of NH_3 -data, namely equal excitation temperature in the hyperfine structure, is not valid in the case of anomalies. On the other hand, as a result of the present study, hfs-anomalies are a rather common phenomenon for intermediate warm clouds. The emission of such clouds therefore should be carefully checked for possible anomalies, before applying the standard NH_3 -data analysis to these sources. In the case of anomalies, a more detailed interpretation with the help of radiative transfer calculations will be necessary (see Stutzki, 1984; Stutzki and Winnewisser, 1984).

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